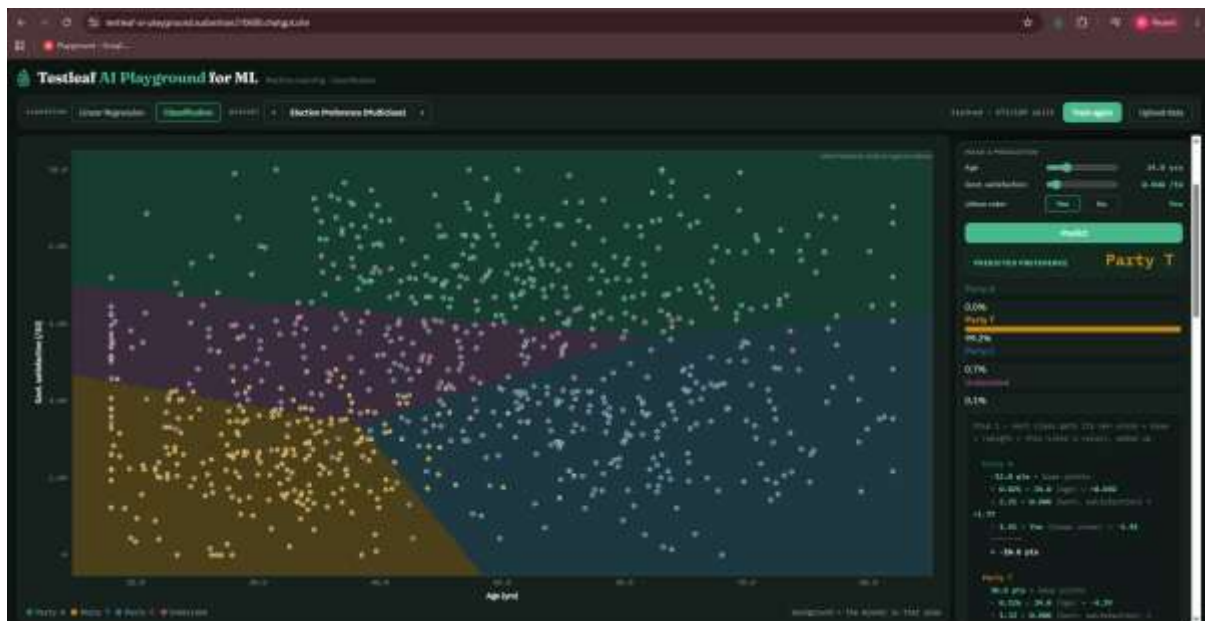


Classification Algorithm

a) Age <30 and Govt satisfaction less and urban



```
Party A
-11.8 pts + base points
+ 0.037 × 34.0 (Age) = +1.27
+ 2.08 × 0.800 (Govt. satisfaction) =
+1.66
- 1.55 × Yes (Urban voter) = -1.55
= -10.5 pts

Party T
10.9 pts + base points
- 0.114 × 34.0 (Age) = -3.86
- 1.46 × 0.800 (Govt. satisfaction) =
-1.17
+ 0.610 × Yes (Urban voter) = +0.610
= 6.47 pts

Party C
0.193 pts + base points
+ 0.114 × 34.0 (Age) = +3.86
- 0.873 × 0.800 (Govt. satisfaction) =
-0.699
- 1.82 × Yes (Urban voter) = -1.82
= 1.54 pts

Undecided
0.710 pts + base points
- 0.011 × 34.0 (Age) = -0.360
+ 0.388 × 0.800 (Govt. satisfaction) =
+0.310
- 0.873 × Yes (Urban voter) = -0.873
= -0.214 pts

highest score wins → Party T
```

Step 2a – convert each score to e^{score}
(always positive)

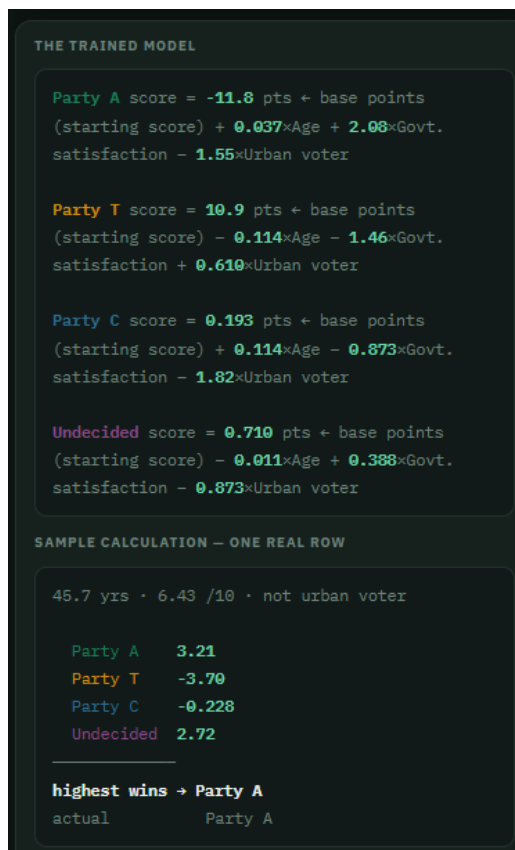
```
Party A     $e^{-10.5} = 0.000$ 
Party T     $e^{6.47} = 645$ 
Party C     $e^{1.54} = 4.64$ 
Undecided   $e^{-0.214} = 0.808$ 
```

Total (add all 4 up) = 651

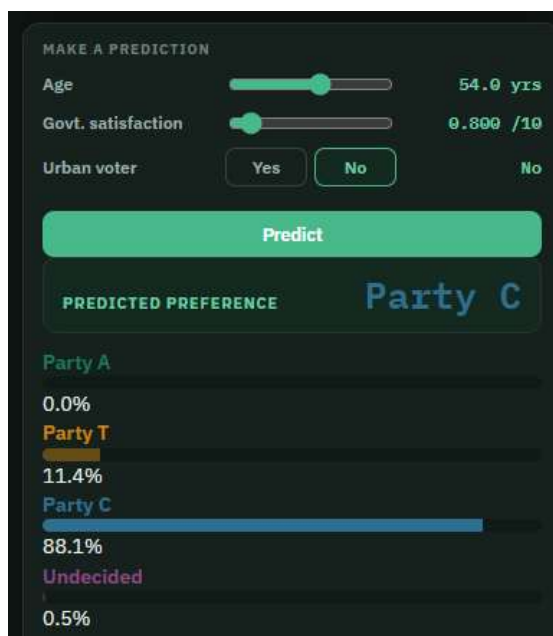
Step 2b – each class's share = its own e^{score}
÷ that same total (651)

```
Party A     $0.000 \div 651 = 0.0\%$ 
Party T     $645 \div 651 = 99.2\%$ 
Party C     $4.64 \div 651 = 0.7\%$ 
Undecided   $0.808 \div 651 = 0.1\%$ 
```

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.



B) Age >50 and Govt satisfaction less , non-urban



Step 1 – each class gets its own score = base + (weight × this voter's value), added up

Party A
 -11.8 pts + base points
 + 0.037 × 54.0 (Age) = +2.01
 + 2.08 × 0.800 (Govt. satisfaction) = +1.66
 - 1.55 × No (Urban voter) = +0
 = -8.17 pts

Party T
 10.9 pts + base points
 - 0.114 × 54.0 (Age) = -6.14
 - 1.46 × 0.800 (Govt. satisfaction) = -1.17
 + 0.610 × No (Urban voter) = +0
 = 3.59 pts

Party C
 0.193 pts + base points
 + 0.114 × 54.0 (Age) = +6.14
 - 0.873 × 0.800 (Govt. satisfaction) = -0.699
 - 1.82 × No (Urban voter) = +0
 = 5.63 pts

Undecided
 0.710 pts + base points
 - 0.011 × 54.0 (Age) = -0.571
 + 0.388 × 0.800 (Govt. satisfaction) = +0.310
 - 0.873 × No (Urban voter) = +0
 = 0.448 pts

Step 2a – convert each score to e^{score} (always positive)

Party A $e^{-8.17} = 0.000$
 Party T $e^{3.59} = 36.2$
 Party C $e^{5.63} = 279$
 Undecided $e^{0.448} = 1.57$

Total (add all 4 up) = 317

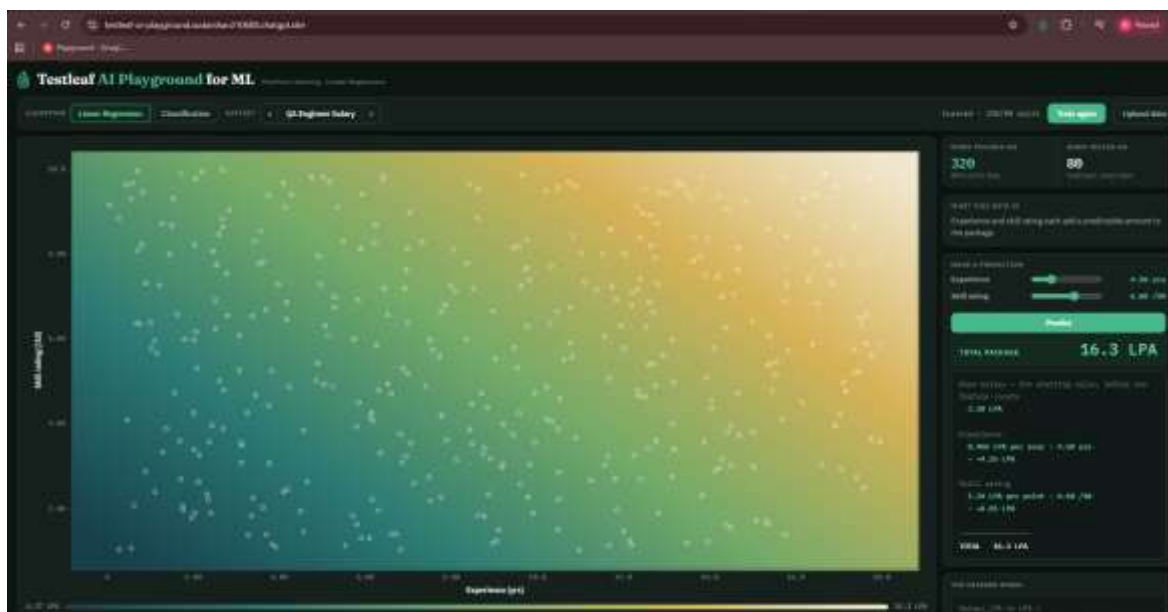
Step 2b – each class's share = its own e^{score} ÷ that same total (317)

Party A $0.000 \div 317 = 0.0\%$
 Party T $36.2 \div 317 = 11.4\%$
 Party C $279 \div 317 = 88.1\%$
 Undecided $1.57 \div 317 = 0.5\%$

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

2)Linear regression

a) QA Salary



b) Food Delivery time

